

MOUHSINE CHRINBOU

Senior Embedded Software Engineer

Embedded Systems | Modern C++ | Embedded Linux | Python | RTOS | Railway | Automotive | Energy

Versailles, France

Email: chrinbou.mouhsine@gmail.com

Professional Summary

Senior Embedded Software Engineer with over 6 years of experience designing and developing high-performance software solutions in safety-critical and industrial environments. Strong expertise in Modern C/C++, Python, Embedded Linux, RTOS, REST APIs, AUTOSAR and distributed software architectures. Proven experience in Energy, Railway, Automotive and IoT industries.

Core Competencies

Programming: C, Modern C++, Python, Bash, SQL, Qt/QML

Embedded: Embedded Linux, STM32, FreeRTOS, UART, SPI, I2C, ADC

Architecture: SOLID, OOP, REST APIs, UML, Design Patterns, Distributed Systems

DevOps: Git, GitLab CI, Jenkins, Docker, Podman, Conan, CMake, Artifactory

Databases: PostgreSQL, MySQL, Oracle, SQL Server

Protocols: TCP/IP, UDP, Kafka, WebSocket, XML, JSON

Tools: VS Code, Qt Creator, Jira, Confluence, PlantUML, Swagger, Gerrit

Professional Experience

Senior Software Engineer – ENEDIS

May 2024 – Present

Designed scalable RESTful services following SOLID principles.

Developed backend services in Modern C++.

Implemented REST APIs, Kafka messaging and WebSocket notifications.

Developed SQL persistence layer and Shell automation.

Implemented unit and integration tests.

Maintained CI/CD pipelines with GitLab CI, Conan and CMake.

Produced architecture documentation using PlantUML and Swagger.

Technologies: Modern C++, Linux, Docker, Kafka, GitLab CI, Conan, CMake, PostgreSQL

Embedded Software Engineer – Hitachi Rail (Segula Technologies)

Jun 2023 – Mar 2024

Developed safety-critical railway software compliant with EN50128.

Designed software using UML.

Implemented TCP/IP and IPC communication.

Developed unit tests and validation activities.

Technologies: C/C++, Linux, TCP/IP, XML, JSON, Git, CMake, SCADE

Embedded Software Engineer – Capgemini Engineering / Dassault Systèmes

Jun 2022 – May 2023

Developed AUTOSAR software components.

Configured BSW/RTE using Vector DaVinci.

Validated software with CANoe.

Developed Python/PyQt tools for ADAS simulation analysis.

Technologies: AUTOSAR, C, Python, PyQt, CANoe, ISO26262

Software Engineer – M2M Group

Dec 2021 – May 2022

Maintained payment software platform.

Investigated and fixed C/C++ issues.

Automated build and deployment with CMake and Shell.

Technologies: C/C++, Oracle, AIX, CMake

Embedded Software Engineer – DIGITORY

Aug 2020 – Oct 2021

Developed STM32 embedded drivers.

Built FreeRTOS applications.

Developed LoRa gateway and IoT connectivity.

Technologies: STM32, FreeRTOS, LoRa, Embedded C

Education

Engineering Degree – Electrical and Embedded Systems Engineering (2017–2020)

Preparatory Classes – Industrial Sciences (2014–2016)

Languages

French (Fluent)

English (Professional)

Arabic (Native)